

HONGJING ZHU | RESUME

Shenzhen Technology University, Guangdong, China

✉ 202400501018@stumail.sztu.edu.cn

☎ (+86) 15107542067

GitHub: 33M4Ximilian



SUMMARY

Undergraduate robotics researcher focusing on robot learning, dexterous manipulation, and embodied AI. Developing reinforcement-learning policies and perception-control systems that bridge simulation, scalable training, and real-robot deployment. Experience spans in-hand manipulation, dual-arm dynamic handover, vision-language-action training infrastructure, and mobile manipulation.

RESEARCH INTERESTS

- Q Embodied AI and Artificial Intelligence
- Q Vision-Language-Action (VLA) Models and World-Action-Model (WAM)
- Q Reinforcement Learning
- Q Dexterous Manipulation

EDUCATION

Shenzhen Technology University | B.Eng. Candidate, Unmanned Aircraft Systems Engineering

Sep 2024 - Present

GPA: 3.0/4.0 | Research interests: Dexterous Manipulation, Robot Learning, Reinforcement Learning, Sim-to-Real, and VLA/VLN.

INTERNSHIP EXPERIENCE

BrainCo | Dexterous Hand Algorithm Intern

Feb 2026 - Jun 2026

Open-source contribution: [RevoLab](#) (RL / Dexterous Manipulation)

➤ RMA In-Hand Rotation

- Adapted the HORA framework to the Revo3 21-DoF dexterous hand.
- Designed a 141-dimensional observation space, 30-step observation history, reward functions, and dynamics randomization.
- Completed two-stage PPO training using the Teacher Policy and ProprioAdapt framework.

➤ Sim-to-Real Deployment

- Built a complete deployment pipeline from PyTorch/ONNX models to the Revo3 Python SDK.
- Implemented observation history reconstruction, policy-to-joint mapping, joint limit handling, offset calibration, and MIT-mode motor control.
- Demonstrated the deployed policy at the BrainCo booth during ICRA 2026.

➤ Dynamic Handover

- Reproduced and adapted the Dynamic Handover framework to the RevoTron dual-arm dexterous manipulation platform.
- Formulated bidirectional object tossing as a command-conditioned POMDP.
- Designed four action groups, stage-wise reward functions, reset/termination logic, and Curriculum Command Distribution (CCD).
- Trained policies using RL-Games PPO.

➤ VLA Training Infrastructure

- Unified LeRobot datasets and state/action schemas.
- Established local training and evaluation pipelines for ACT, $\pi 0.5$, and NVIDIA GR00T N1.6.
- Completed LoRA fine-tuning and multi-GPU adaptation on RTX 5090 GPUs.

RESEARCH EXPERIENCE

2025 AI Robot Simulation Development Challenge | Mobile Manipulation System

Nov 2025 - Jan 2026

First Prize, Undergraduate Division | Ranked 2nd in preliminaries and 1st in finals

- **Mobile Manipulation**
 - Integrated Isaac Sim, ROS 2, SLAM Toolbox, Nav2, stereo AprilTag perception, and MoveIt.
 - Developed a complete navigation–perception–grasping–placement pipeline.
- **6D Perception & Control**
 - Implemented stereo triangulation, SolvePnP, and TF-based pose estimation.
 - Coordinated NavigateToPose, MoveGroup, and gripper execution through an asynchronous state machine.

Robot Vision & 6D Pose Estimation | RealSense + GraspNet / FoundationPose

Aug 2025 - Nov 2025

- **GraspNet**
 - Implemented RGB-D filtering, point cloud generation, grasp pose refinement and tracking.
 - Published end-effector targets via ROS 2 and executed grasping using MoveIt inverse kinematics.
- **FoundationPose**
 - Built a perception pipeline using YOLO and SAM for object detection and segmentation.
 - Generated object poses and implemented camera-arm calibration, coordinate transformations, depth noise handling, and perception-to-execution interfaces.

CORE SKILLS

- Reinforcement Learning / Dexterous Manipulation:**
POMDP task formulation、PPO、Observation/action/reward design、Dynamics randomization、ProprioAdapt
- VLA / Imitation Learning:**
ACT、 $\pi 0.5$ 、NVIDIA GR00T N1.6、LeRobot、LoRA fine-tuning
- Simulation & Deployment:**
Isaac Sim、Isaac Lab、RSL-RL、RL-Games、PyTorch、ONNX
- Robotics Systems:**
ROS 2、SLAM Toolbox、Nav2、MoveIt 2、YOLO、SAM

HONORS & AWARDS

2025 AI Robot Simulation Development Challenge | Undergraduate First Prize

Jan 2026

Lanqiao Cup Python Programming Competition | Provincial Second Prize

May 2025